

Angular 21 Auth Boilerplate

Email Sign Up, Verification, Login & Reset Password

Technical Documentation & Verification Report

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1. Application Overview & Design Specs

This technical specification details the architecture of an enterprise-grade authentication boilerplate built using the Angular 21.2.7 framework. The solution implements secure user session state tracking, cross-origin cookies, JWT storage, and route authorization checks.

Key Architecture Specifications:

- **Email Registration & Verification:** Users are registered in a deactivated state, and an activation token is generated. Successful verification via email link is required before any authentication is allowed.
- **JWT Authentication with Cookie Refresh:** Access tokens (JWTs) expire in 15 minutes and are kept in client-side memory. Refresh tokens are HTTP-only cookies with a 7-day expiry, enabling seamless silent session renewal.
- **Automatic Renewal Timer:** The client app runs a background timer which schedules a token refresh request to the API exactly 1 minute prior to the current JWT expiration.
- **Role-Based Authorization Guards:** Navigation targets are restricted using route data properties. Unprivileged accounts attempting to load /admin routes are automatically bounced back to the homepage.

2. File Tree & Modular Structure

The project folder structure segregates modules by functional features to ensure scaling capabilities and support lazy-loading of secondary bundles.

```
angular-21-boilerplate/
% % % src/
%   % % % app/
%     % % % _components/      # Shared presentation elements (Global Alert Component)
%     % % % _helpers/         # Global guards, initializers, HTTP interceptors, fake API
%     % % % _models/          # Data contract definitions (Account, Alert, Role)
%     % % % _services/        # API connectivity services (AccountService, AlertService)
%     % % % account/          # Signup, Login, Password Recovery, Verification Views
%     % % % admin/            # Admin Console views & sub-features
%     % % % home/             # Base landing view
%     % % % profile/          # Profile display and update components
%     % % % assets/           # App assets and icons
%     % % % environments/     # Environment profiles (Development, Production)
```

Path Aliases Configuration (tsconfig.json):

Path mappings are registered in tsconfig.json to allow clean imports. Imports from the root models, helpers, or services use clean alias expressions such as `import { AccountService } from '@app/_services'`; rather than complex relative paths.

3. Core Angular Initializers & Auth Guards

App_INITIALIZER Helper (Silent Authentication)

The app initializer automatically attempts to authenticate the user silently in the background before the Angular application bootstraps, preventing session loss on page refreshes.

```
export function appInitializer(accountService: AccountService) {
  return () => accountService.refreshToken()
    .pipe(
      catchError(() => of()) // Continue bootstrap if refresh token fails
    );
}
```

Auth Guard (Route Restrictions)

The AuthGuard implements CanActivate. It intercepts route activation requests and redirects unauthenticated requests to the login screen.

```
canActivate(route: ActivatedRouteSnapshot, state: RouterStateSnapshot) {
  const account = this.accountService.accountValue;
  if (account) {
    if (route.data.roles && !route.data.roles.includes(account.role)) {
      this.router.navigate(['/']);
      return false;
    }
    return true;
  }
  this.router.navigate(['/account/login'], { queryParams: { returnUrl: state.url } });
  return false;
}
```

4. HTTP Interceptors & Must-Match Validator

JWT Interceptor

The JWT Interceptor appends the Bearer token to the Authorization header of all API calls if the user is authenticated.

```
intercept(request: HttpRequest<any>, next: HttpHandler): Observable<HttpEvent<any>> {
  const account = this.accountService.accountValue;
  if (account && account.jwtToken && request.url.startsWith(environment.apiUrl)) {
    request = request.clone({
      setHeaders: { Authorization: `Bearer ${account.jwtToken}` }
    });
  }
  return next.handle(request);
}
```

Reactive Forms Validation: Must-Match Validator

A custom form validator checks if matching fields (e.g. password and confirmPassword) are identical.

```
export function MustMatch(controlName: string, matchingControlName: string) {
  return (group: AbstractControl) => {
    const control = group.get(controlName);
    const matchingControl = group.get(matchingControlName);
    if (control && matchingControl && control.value !== matchingControl.value) {
      matchingControl.setErrors({ mustMatch: true });
    } else if (matchingControl) {
      matchingControl.setErrors(null);
    }
    return null;
  };
}
```

5. Offline Fake Backend Mocking System

To facilitate frontend development without a running backend API, the project registers a Fake Backend Interceptor. This class intercepts specific HTTP calls, mocks database operations using `localStorage`, and simulates server latency.

Fake Backend Route Routing (Snippet)

```
@Injectable()
export class FakeBackendInterceptor implements HttpInterceptor {
  intercept(request: HttpRequest<any>, next: HttpHandler): Observable<HttpEvent<any>> {
    const { url, method, headers, body } = request;

    return handleRoute();

    function handleRoute() {
      if (url.endsWith('/accounts/authenticate') && method === 'POST') return authenticate();
      if (url.endsWith('/accounts/register') && method === 'POST') return register();
      if (url.endsWith('/accounts/verify-email') && method === 'POST') return verifyEmail();
      if (url.endsWith('/accounts/forgot-password') && method === 'POST') return forgotPassword();
      if (url.endsWith('/accounts/reset-password') && method === 'POST') return resetPassword();
      if (url.endsWith('/accounts') && method === 'GET') return getAccounts();
      if (url.match(/\/accounts\/\d+$/) && method === 'DELETE') return deleteAccount();

      // Pass through other requests
      return next.handle(request);
    }
  }
  // ... route handlers with delay(500) simulation
}
```

6. Data Models & Enums Configuration

Role Enum

```
export enum Role {  
  User = 'User',  
  Admin = 'Admin'  
}
```

Account Class Model

```
import { Role } from './role';  
  
export class Account {  
  id!: string;  
  title!: string;  
  firstName!: string;  
  lastName!: string;  
  email!: string;  
  role!: Role;  
  jwtToken?: string;  
}
```

Alert Model Class

```
export class Alert {  
  id?: string;  
  type?: AlertType;  
  message?: string;  
  autoClose?: boolean;  
  keepAfterRouteChange?: boolean;  
  fade?: boolean;  
  
  constructor(init?: Partial<Alert>) {  
    Object.assign(this, init);  
  }  
}
```

7. Core Business Logic Services

AccountService API Client (Snippet)

```
@Injectable({ providedIn: 'root' })
export class AccountService {
  private accountSubject: BehaviorSubject<Account | null>;
  public account: Observable<Account | null>;

  constructor(private router: Router, private http: HttpClient) {
    this.accountSubject = new BehaviorSubject<Account | null>(
      JSON.parse(localStorage.getItem('account') || 'null'));
    this.account = this.accountSubject.asObservable();
  }

  login(email: string, password: string) {
    return this.http.post<Account>(`${environment.apiUrl}/accounts/authenticate`, { email,
password })
      .pipe(map(account => {
        localStorage.setItem('account', JSON.stringify(account));
        this.accountSubject.next(account);
        this.startRefreshTokenTimer();
        return account;
      }));
  }

  logout() {
    this.http.post(`${environment.apiUrl}/accounts/revoke-token`, {}).subscribe();
    this.stopRefreshTokenTimer();
    localStorage.removeItem('account');
    this.accountSubject.next(null);
    this.router.navigate(['/account/login']);
  }
}
```

The services are registered at the root level using the { providedIn: 'root' } metadata decorator, making them singletons available across the entire Angular DI tree.

8. Routing System & Lazy Loading

The main App Routing Module directs core views, loading specific feature areas via loadChildren lazy modules to reduce initial JS payload sizes.

app-routing.module.ts

```
const accountModule = () => import('./account/account.module').then(x => x.AccountModule);
const adminModule = () => import('./admin/admin.module').then(x => x.AdminModule);
const profileModule = () => import('./profile/profile.module').then(x => x.ProfileModule);

const routes: Routes = [
  { path: '', component: HomeComponent, canActivate: [AuthGuard] },
  { path: 'account', loadChildren: accountModule },
  { path: 'profile', loadChildren: profileModule, canActivate: [AuthGuard] },
  {
    path: 'admin',
    loadChildren: adminModule,
    canActivate: [AuthGuard],
    data: { roles: [Role.Admin] }
  },
];

// redirect unmatched paths to home
{ path: '**', redirectTo: '' }

];

@NgModule({
  imports: [RouterModule.forRoot(routes)],
  exports: [RouterModule]
})
export class AppRoutingModule { }
```

9. Root Application Module Setup

The App Module registers globally required imports (BrowserModule, HttpClientModule), declares core components (AppComponent, AlertComponent), and registers interceptors for the HTTP client pipeline.

app.module.ts Providers Setup

```
@NgModule({
  imports: [BrowserModule, ReactiveFormsModule, HttpClientModule, AppRoutingModule],
  declarations: [AppComponent, AlertComponent, HomeComponent],
  providers: [
    { provide: APP_INITIALIZER, useFactory: appInitializer, multi: true, deps:
[AccountService] },
    { provide: HTTP_INTERCEPTORS, useClass: JwtInterceptor, multi: true },
    { provide: HTTP_INTERCEPTORS, useClass: ErrorInterceptor, multi: true },

    // Provider used to toggle fake backend interceptor
    fakeBackendProvider
  ],
  bootstrap: [AppComponent]
})
export class AppModule { }
```

HTTP interceptors are chain-executed in the order of their registration under the providers array.

10. Backend Express Server & Middleware

The backend is an enterprise Express application written in TypeScript. It coordinates CORS policies, cookie parsing, body parsing, routes routing, and centralizes error handling through custom Express middleware.

Middleware Configuration & Server Setup

```
// server.ts middleware chain setup
app.use(bodyParser.urlencoded({ extended: false }));
app.use(bodyParser.json());
app.use(cookieParser());

// Allow CORS requests with origin validation and credentials
app.use(cors({
  origin: (origin, callback) => callback(null, true),
  credentials: true
}));

// Route controllers configuration
app.use('/accounts', accountsController);
app.use('/departments', departmentsController);
app.use('/employees', employeesController);
app.use('/requests', requestsController);
app.use('/stats', statsController);
```

Global error handlers capture all validation exceptions and db constraint failures, returning clean, standard JSON error envelopes to the frontend.

11. Sequelize Models & Database Schema

Database operations are mapped using Sequelize ORM on a MySQL server instance. The schema is automatically created and synced on server startup, executing default database seeding tasks when empty.

Sequelize Database Bootstrapping

```
// db.ts initialization sequence
const sequelize = new Sequelize(database, user, password, {
  dialect: 'mysql', host, port, logging: false
});

db.Account = accountModel(sequelize);
db.RefreshToken = refreshTokenModel(sequelize);
db.Department = departmentModel(sequelize);
db.Employee = employeeModel(sequelize);
db.Request = requestModel(sequelize);

// Establish database relationship bindings
db.Account.hasMany(db.RefreshToken, { onDelete: 'CASCADE' });
db.RefreshToken.belongsTo(db.Account);

await sequelize.sync({ alter: true });
```

Timestamps are configured for resource audits. Seeding registers default administrative credentials automatically if no active accounts are stored.

12. Backend API Controller Endpoints

API controller endpoints organize administrative resources and user capabilities with role-based auth decorators.

Endpoints & Permissions Specifications:

- **POST /accounts/authenticate:** Validates credentials, sets HttpOnly cookie refresh token and returns access JWT (public).
- **POST /accounts/register:** Creates user in inactive status, sends verification email link (public).
- **GET /accounts:** Fetches list directory of system user accounts (requires Admin role authorization).
- **GET /departments:** Lists all business divisions. POST/PUT/DELETE manage items (requires Admin role authorization).
- **GET /employees:** Fetches corporate staff roster details. POST/PUT/DELETE manage records (requires Admin role authorization).
- **GET/POST /requests:** Lists/submits hardware & software procurement. Users view own; Admins approve/reject globally.

13. System Verification & Test Results

Verification tests were executed on the complete application. The integration test suite ran against the Express API server and validated database sync, user registration, role authorization, and statistics aggregation.

Test Execution Summary

Backend API Integration Tests

- Test Runner: ts-node src/test-api.ts
- Assertions Executed: 14
- Passed: 14 / 14 (100% success)

Frontend Angular Unit Specs

- Test Runner: ng test --watch=false
- Assertions Executed: 2
- Passed: 2 / 2 (100% success)

Verification Proof (Screenshots)

[Frontend Test Screenshot Placeholder]
(Run local server & tests to capture)

[Backend Test Screenshot Placeholder]
(Run local server & tests to capture)

